Assignment no = 9

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1) Selection Sort :—

{ 18, 22, 20, 19, 21 } = unsorted.

find ~~min~~ minimum element from unsorted array and swap with first element of unsorted array.

* min = 18.
Swap with unsorted = 12.

{ 18, 22, 20, 19, 21 }
Sorted

min = 19
Swap with 22.
{ 18, 19, 20, 22, 21 }
Sorted.

min = 20
Swap with 20.
{ 18, 19, 20, 22, 21 }
Sorted.

min = 21
Swap with 22.
{ 18, 19, 20, 21, 22 }
Sorted.

remaining only One.
So array is Sorted.

{ 18, 19, 20, 21, 22 }

~~# Code~~

2) {34, 12, 56, 45, 23} = Unsorted.

min = 12
Swap with 34.

{12, 34, 56, 45, 23}
sorted

min = 23
Swap with 34.
{12, 23, 56, 45, 34}
Sorted.

min = 34.
Swap with 56.
{12, 23, 34, 45, 56}
Sorted.

min = 45.
Swap with 45.
{12, 23, 34, 45, 56}
sorted.

{12, 23, 34, 45, 56} - Sorted.

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#

```
                    0  1  2  3  4
int arr[] = { 18, 22, 20, 19, 21 }
for (int i = 0; i < arr.length - 1; i++) {
        int minIndex = i;
        for (int j = i+1; j < n; j++) {
                if (arr[j] < arr[minIndex]
                        minIndex = j;
                }
        }
        int temp = arr[i];
        arr[i] = arr[minIndex]
        arr[minIndex] = temp;
}
```

Dryrun :- : arr.length = 5.

i = 0 < 5 - 1 = 4
└→ min = 0 — j = 1 < 5
└ 22 < 18 — X.
j = 2 < 5
└ arr[2] = 20 < 18 — X
j = 3 < 5
└ arr[3] = 19 < 18 — X
j = 4 < 5
└ 21 < 18 — X.

temp = 18;
18 = 18 = no swap.
18 = 18.
i++.

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{18, 22, 20, 19, 21}

i = 1 < 5 - 1 = 4
└ min = 1 = j = 2 < 5
└ 20 < 22 - true
└ min = 2
j = 3 < 5
└ 19 < 20 - true
└ min = 3
j = 4 < 5
└ 21 < 19 - X.

temp = 22
22 = ~~20~~ 19
~~20~~ 19 = 22

{18, 19, 20, 22, 21}
0 1 2 3 4

i = 2 < 5 - 1 = 4
└ min = 2 = j = 3 < 5
└ ~~19~~ 22 < 20 - X
j = 4 < 5
└ 21 < 20 - X.

no swap.

i = 3 < 5 - 1 = 4
└ min = 3 = j = 4 < 5
└ 21 < 22 - ✓ = min = 4

temp = 22.
22 = 21
21 = 22

{18, 19, 20, 21, 22}

swap हो गया.

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Code = int arr[]={34, 12, 56, 45, 23}

```
for (int i=0; i< arr.length-1; i++){
    int minIndex = i;
    for (int j= i+1; j<n; j++){
        if (arr[j] < arr[minIndex]){
            minIndex = j;
        }
    }
    int temp = arr[i];
    arr[i] = arr[minIndex];
    arr[minIndex = temp;
}
```

+ Dryrun :-

i=0<4 — min = 0 = j=1<5
└ 12<34 ✓
└ min=1
j=2<5
└ 56<12 ✗
j=3<5
└ 45<12 ✗
j=4<5
└ 23<12 ✗

temp = 34
34 = 12 Swap.
12 = 34 temp.

{12, 34, 56, 45, 23}
0 1 2 3 4

$i = 1 < 5-1 = 4 = min = 1 = j = 2 < 5$

$\quad$ 56 < 34 ✗

$j = 3 < 5$

$\quad$ 45 < 34 ✗

$j = 4 < 5$

$\quad$ 23 < 34 ✓

$\quad$ min = 4

temp = 34

34 = 23 $\quad$ Swap

23 = temp

$\{ \overset{0}{12}, \overset{1}{23}, \overset{2}{56}, \overset{3}{45}, \overset{4}{34} \}$

$i = 2 < 5-1 = 4 = min = 2 = j = 3 < 5$

$\quad$ 45 < 56 ✓

$\quad$ min = 3

$j = 4 < 5$

$\quad$ 34 < 45 ✓

$\quad$ min = 4

temp = 56.

56 = 34 $\quad$ Swap.

34 = temp

$\{ \overset{0}{12}, \overset{1}{23}, \overset{2}{34}, \overset{3}{45}, \overset{4}{56} \}$

$i = 3 < 5-1 = 4 = min = 3 = j = 4 < 5$

$\quad$ 56 < 45 ✗

$i = 4 < 4$ ✗. array is

$\{ 12, 23, 34, 45, 56 \}$

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2) Bubble Sort :-

{30, 25, 27, 35, 29} = unsorted.

{30, 25, 27, 35, 29} 30 > 25 Swap.

{25, 30, 27, 35, 29} 30 > 27 Swap.

{25, 27, 30, 35, 29} 30 > 35 ×.

{25, 27, 30, 35, 29} 35 > 29 - Swap.

{25, 27, 30, 29, 35} Step 1 Complete.
Sorted.

{25, 27, 30, 29, 35} 30 > 29. Swap.

{25, 27, 29, 30, 35}
Sorted.

{25, 27, 29, 30, 35}

{25, 27, 29, 30, 35}
Sorted.

{25, 27, 29, 30, 35} Sorted.

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{34, 12, 56, 45, 23 }. unsorted.

{34, 12, 56, 45, 23 } 34>12 swap.

{12, 34, 56, 45, 23 } 56>45 swap.

{12, 34, 45, 56, 23 } 56>23 swap.

{12, 34, 45, 23, 56 }
sorted.

{12, 34, 45, 23, 56 } 45>23 . swap.

{12, 34, 23, 45, 56 }
Sorted.

{12, 34, 23, 45, 56 } 34>23 swap.

{12, 23, 34, 45, 56 }
sorted.

{12, 23, 34, 45, 56 }
Sorted.

{12, 23, 34, 45, 56 } Sorted.

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Code:

```
int arr[] = { 30, 25, 27, 35, 29 }
for (int i = 0; i<n-1 ; j++) {
    for (int j = 0 ; j< n-i-1 ; j++){
        if (arr[j] > arr[j+1])
            int temp = arr[j]
            arr[j] = arr[j+1]
            arr[j+1] = temp;
        }
    }
}
```

Dryrun:-

$i = 0 < 4$

$\quad$ ∟ $j = 0 < n-i-1 = 4$ $\quad$ (5-0-1)

$\quad\quad$ ∟ $30 > 25$. - swap

$\quad\quad\quad$ ∟ temp = 30

$\quad\quad\quad$ 30 = 25

$\quad\quad\quad$ 25 = temp

$\{ 25, 30, 27, 35, 29 \}$

$j = 1 < 4$

$\quad$ ∟ $30 > 27$ - swap.

$\quad\quad$ ∟ temp = 30

$\quad\quad$ 30 = 27

$\quad\quad$ 27 = temp.

$\{ 25, 27, 30, 35, 29 \}$

$j = 2 < 4$

$\quad$ ∟ $30 > 35$ ✗.

$j = 3 < 4$

$\quad$ ∟ $35 > 29$ — swap.

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$\{ \overset{0}{25}, \overset{1}{27}, \overset{2}{30}, \overset{3}{29}, \overset{4}{35} \}$

$i = 1 < 4$

$\quad j = 0 < 5-1-1 = 3$

$\quad\quad 25 > 27$ ✗

$\quad j = 1 < 4$

$\quad\quad 27 > 30$ ✗

$\quad j = 2 < 4$

$\quad\quad 30 > 29$ — swap.

$\{ \overset{0}{25}, \overset{1}{27}, \overset{2}{29}, \overset{3}{30}, \overset{4}{35} \}$

$\quad j = 3 < 4$

$\quad\quad 30 > 35$ ✗.

$i = 2 < 4$

$\quad j = 0 < 5-2-1 = 2$.

$\quad\quad 25 > 27$ ✗

$\quad j = 1 < 4$

$\quad\quad 27 > 29$ ✗

$\quad j = 2 < 4$

$\quad\quad 29 > 30$ ✗

$\quad j = 3 < 4$

$\quad\quad 30 > 35$ ✗.

$i = 3 < 4$

$\quad j = 0 < 4$

$\quad\quad 25 > 27$ ✗

$\quad j = 1 < 4$

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Code:

```
int arr[] = { 34, 12, 56, 45, 23 }
for (int i=0; i<n-1; i++) {
    for (int j=0; j<n-i-1; j++) {
        if (arr[j] > arr[j+1]) {
            int temp = arr[j];
            arr[j] = arr[j+1];
            arr[j+1] = temp;
        }
    }
}
```

Dryrun:-

i = 0 < 4

└ j = 0 < 4

└ 34 > 12 ✓ swap.

{ 12, 34, 56, 45, 23 }

j = 1 < 4

└ 34 > 56 ✗

j = 2 < 4

└ 56 > 45 ✓ swap.

{ 12, ~~56~~, 34, 45, 56, 23 }

j = 3 < 4

└ 56 > 23 ✓ swap.

12, 34, 45, 23, 56 }

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$i = 1 < 4$

└ $j = 0 < 3$

└ $12 > 34$ ✗

$j = 1 < 3$

└ $34 > 45$ ✗

$j = 2 < 3$

└ $45 > 23$ ✓ swap.

$\{ \overset{0}{12}, \overset{1}{34}, \overset{2}{23}, \overset{3}{45}, \overset{4}{56} \}$

$i = 2 < 4$

└ $j = 0 < 2$

└ $12 > 34$ ✗

$j = 1 < 2$

└ $34 > 23$ ✓ swap.

$\{ \overset{0}{12}, \overset{1}{23}, \overset{2}{34}, \overset{3}{45}, \overset{4}{56} \}$

$i = 3 < 4$

└ $j = 0 < 1$

└ $12 > 23$ ✗.

array is = $\{ 12, 23, 34, 45, 56 \}$
Sorted.

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3) insertion Sort :-

{18, 22, 20, 19, 21} - Consider 18 sorted.
Sorted

Current
{18, 22, 20, 19, 21} Compare 22 & 18.
22 > 18. then no shift.
{18, 22, 20, 19, 21} ~~insert~~.
Sorted

{18, 22, 20, 19, 21} left = 22 right = 20 22>20
shift.
{18, 22, 22, 19, 21} then insert.

{18, 20, 22, 19, 21} left = 22 right = 19 22>19
Sorted. Shift.

{18, 20, 22, 22, 21} left = 20 right = 19 20>19
shift.

{18, 20, 20, 22, 21} then insert.

{18, 19, 20, 22, 21} left = 22 right = 21 22>21
Sorted. Shift.

{18, 19, 20, 22, 22} left 20 right = 21 20>21
no shift. insert.

{18, 19, 20, 21, 22} - Sorted.

2) {34, 12, 56, 45, 23} if 34 is sorted.

{34, 12, 56, 45, 23} 34 > 12 then shift.

{34, 34, 56, 45, 23} insert.

{12, 34, 56, 45, 23} 56 < 34 no shift.
Sorted.

{12, 34, 56, 45, 23} 56 > 45 then shift.
Sorted.

{12, 34, 56, 56, 23} then insert.

{12, 34, 45, 56, 23} 56 > 23 shift.
Sorted.

{12, 34, 45, 56, 56} left = 45 right = 23 45 > 23
shift.

{12, 34, 45, 45, 56} left = 34 right = 23 34 > 23
shift.

{12, 34, 34, 45, 56} insert.

{12, 23, 34, 45, 56} Sorted.

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Code :

```
int arr[] = { 18, 22, 20, 19, 21 }
for (int i = 1 ; i < 5 ; i++) {
    int current = ~~i++~~ arr[i];
    int left = i-1;
    while (left > 0 && arr[left] > current)
        arr[left+1] = arr[left];
        left--;
    }
    arr[left+1] = current;
}
```

Dryrun :-

i = 1 < 5

└ current = ~~18~~ 22 left = 0

└ ~~while~~ 0 >= 0 && 18 > ~~18~~ 22 ×

arr[left+1] = arr[1] = 22 = 18

{ 18, 22, 20, 19, 21 }

i = 2 < 5

└ current = 20 left = 1

└ 1 > 0 && 22 > 20

└ 20 = 22 shift.

{ 18, 22, 22, 19, 21 } = ~~arr[2] = 20;~~

left -- ; left = 0

arr[1] = 20 ;

{ 18, 20, 22, 19, 21 }

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i = 3 < 5
└ Current. 19 left = 2
└ 2 > 0 && 22 > 19
└ 19 = 22 , left = 1.
{18, 20, 22, 22, 21} = ~~22 = 19~~.

~~{18~~, Current = 19 , left = 1.
└ 1 > 0 && 20 > 19
└ 22 = 20
left -- = 0

0 1 2 3 4
{18, 20, 20, 22, 21}

└ 0 > 0 && 18 > 19 X.
array is - {18, 19, 20, 22, 21}
0 1 2 3 4

i = 4 < 5
└ Current = 21 left = 3
└ 3 > 0 && 22 > 21
└ 21 = 22, left -- = 2

0 1 2 3 4
{18, 19, 20, 22, 22}

└ 2 > 0 && ~~22~~ 20 > 22 = X.

{18, 19, 20, 21, 22} = Sorted.

array is Sorted.

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Code:

```
int arr[] = {34, 12, 56, 45, 23}
    for (int i=1; i<n; i++){
        int current = arr[i];
        int left = i-1
        while ( left >=0 && arr[left] > current
            arr[left+1] = arr[left];
            left--;
        }
        arr[left+1] = current;
    }
}
```

Dryrun:-

* i=1 < 5
  - C=12, L=0.
    - 0>=0 && 34>12
      - 12 = 34 left=-1

{34, 34, 56, 45, 23}

~~34 = 12~~; arr[0]=12.

{12, 34, 56, 45, 23}

* i=2<5
  - C=56, L=~~34~~ 1
    - 1>0 && 34>56 X

* i=3<5
  - C=45, L=2
    - 2>0 && 56>45
      - 45 = 56.

{12, 34, 56, 56, 23} left -- =1.

arr[1] = 45,

0 1 2 3 4
{12, 34, 45, 56, 23}

i = 4 < 5
└ C = 23, left = 3.
└ 3 > 0 && 56 > 23
└ 23 = 56.
left -- = 2.

0 1 2 3 4
{12, 34, 45, 56, 56}

C = 23, left = 2
└ 2 > 0 && 45 > 23
└ 56 = 45
left -- = 1.

0 1 2 3 4
{12, 34, 45, 45, 56}

C = 23, left = 1
└ 1 > 0 && 34 > 23
└ 45 = 34
left -- = 0.

0 1 2 3 4
{12, 34, 34, 45, 56}

arr[1] = 23.

{12, 23, 34, 45, 56}

i = 5 < 5 = X.

array is : {12, 23, 34, 45, 56}

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* Quick Sort : = {75, 85, 55, 95, 65}

75 pivot.

55	65	75	85	95
0	1	2	3	4

{55, 65} → {55, 65} → {65}

{85, 95} → {85, 95} → {95}

2) {42, 56, 12, 67, 34} 42 = pivot.

12	34	42	56	67
0	1	2	3	4

{12, 34} → {12, 34} → {34}

{56, 67} → {56, 67} → {67}

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Code :-

```
                0   1   2   3   4
int arr[] = { 75, 85, 55, 95, 65 }
    f ( arr, 0, arr.length -1);
}
void f (int arr[], int low int high) {
    if (low == high) {
        return
    }
    int pivot = quick (arr, low, high);
    f ( arr, low, pivot -1);
    f (arr, pivot +1, high);
}
}

int quick ( arr, int low, int high){
    int pivot = arr [low];
    int i = low
    int j = high;
    while (i<j) {
        while (i<=high && arr[i] <=pivot) {
            i++; }
    while ( arr [j] > pivot) { j--}
    if ( i<j) {
        int temp = arr[j];
        arr [i] = arr [j];
        arr [j] = temp;
    }
    int temp = arr [low];
    arr [low] = arr [j];
    arr [j] = temp;
    return j;
}
```

Dry run

{ 75, 85, 55, 95, 65 }

f (arr, 0, 4)
low = 0, high = 4
pivot = 1

quick (0, 4)
i = 0 arr [i] < pivot
j = 4 L 75 <= 75 = i = 1.
arr [j] > pivot
65 > 75 x stop.
i<j = 1 < 4 - swap.

{ 75, 65, 55, 95, 85 }
0 1 2 3 4

i = 1 65 < 75 x i++ i = 2
55 < 75 x i = 3
95 < 75 x Stop.

arr [j] > pivot.
85 > 75 j = 4 - j = 3
95 > 75 j = 2

i = 3 & j = 2 i<j = x Stop.
pivot swap.
temp = 75
75 = arr [2] = 55.
55 = temp.

{ 55, 65, 75, 95, 85 } return j = 2

0 1

f (low , pivot - 1)

{55, 65 } pivot = 55.

3 4

f (pivot + 1 , high)

{95, 85 }

pivot = 95. swap. {85, 95 }

{42, 56, 12, 67, 34}

f(low, high
0 4

pivot = 42.

i = 0

j = 4

0 < 4 → 0 < 4 && 42 <= 42 : i = 1

j = 4 34 > 42 × false.

1 < 4 — swap.

{42, 34, 12, 67, 56}

1 < 4 && 34 <= 42 ✓ i = 2.

2 < 4 && 12 <= 42 ✓ i = 3

3 < 4 && 67 <= 42 × false.

- 56 > 42 ✓ j = 3

  67 > 42 ✓ j = 2

  12 > 42 × false.

3 < 2 × false. pivot swap.

{12, 34, 42, 67, 56} j = 2

f(0, pivot-1) = 1 f(0, 1)

pivot = 12 i = 0 j = 1.

12 <= 12 i = 1 34 > 12 j = 0

34 <= 12 ×

i < j : 1 < 0 × . pivot swap.

{12, 34}

f (pivot+1 , high)
3 4

pivot = 67
i = 3 j = 4 .

67 <= 67 = i = 4
56 <= 67 = i = 5 .

56 > 67 = X j = 4
i < j = 5 < 4 X . pivot swap .

{56, 67}

{12, 34, 42, 56, 67} Array Sorted

4] ~~merge Sort~~ :-

{50, 23, 9, 18, 61, 32}

{50, 23} {9} {18, 61} {32}

{50} {23} {18} {61}

{50} {23} {9} {18} {61} {32}

{23, 50} {9} {18, 61} {32}

{23, 50, 9} {18, 61, 32}

{9, 23, 50} + {18, 32, 61}

{9, 18, 23, 32, 50, 61}

```
// Code = int arr[] ={ 50,23,9,18,61,32}
        for (arr, 0, arr.length -1);
        for (int i=0; i< arr.length; i++){
                Sop (arr[i]);
        }
}

void f ( int arr[], int low, int high){
        if (low>= high){
                return
        }
```

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```
int mid = (low+high)/2;
f(arr, low, mid);
f(arr, mid+1, high);
merged (arr, low, mid, high);
}
static void merged (int arr[], int low, int mid, int high){
    int merged[] = new int [high-low+1];
    int blue = low;
    int green = mid+1;
    int red = 0;
    while (blue <=mid && green <=high){
        if (arr[blue] < arr[green]){
            merged[red] = arr[blue];
            red++;
            blue++;
        }else{
            merged[red] = arr[green];
            red++;
            green++;
        }
    }
    while (blue <=mid){
        merged[red] = arr[blue];
        red++;
        blue++;
    }
    while (green <=high){
        merged[red] = arr[green];
        red++;
        green++;
    }
for (int i=0; i< merged.length; i++){
        arr[low+i] = merged[i];
```

2) {12, 45, 1, 23, 67, 90}

{12, 45, 1} {23, 67, 90}

{12, 45} {1} {23, 67} {90}

{12} {45} {23} {67}

{12} {45} {1} {23} {67} {90}

{12, 45} {1} {23, 67} {90}

{12, 45, 1} {23, 67, 90}

{1, 12, 45} + {23, 67, 90}

{1, 12, 23, 45, 67, 90}